

Process Costing

24 problems · workbook pages 8 to 16

What this module actually is

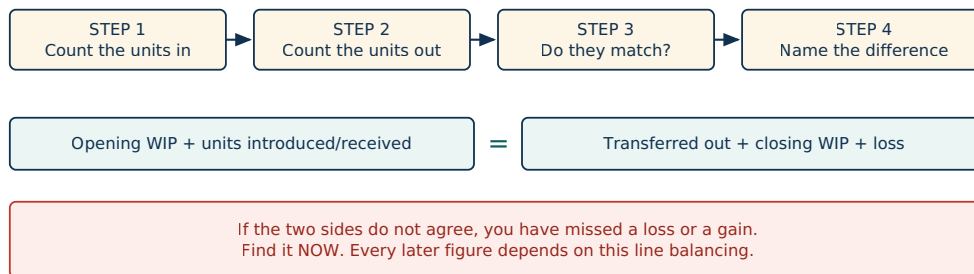
Process costing is used where a product is made in a **continuous chain of stages** — sugar, chemicals, oil, paper, cement. You cannot point at one unit and say “this one cost ₹40”, because every unit goes through the same pipe. So we do the only thing possible: **total the cost of a process and divide by the units it produced**. That single sentence is the whole module.

Everything else in these 24 problems is one of four complications layered on top of that idea:

Complication	What changes	Problems
Nothing — plain chain	Total each process, divide by units, carry the total forward to the next process.	1-4
Some units are lost	Part of the input disappears (evaporation, spoilage). Expected loss is <i>normal</i> and its cost is absorbed by the good units. Anything worse is <i>abnormal loss</i> ; anything better is <i>abnormal gain</i> . Both are taken out of the process at the good-unit rate.	5-9
Two or more products emerge	One process throws out several saleable products. The joint cost has to be split between them — usually by working <i>backwards</i> from selling price.	10-14
Each process adds profit	A process sells to the next at cost + profit. The profit inside unsold stock is not real yet, so a <i>reserve for unrealised profit</i> must be removed.	15-16
Some units are half-finished	Closing work-in-progress is 60% done. You convert part-finished units into <i>equivalent</i> finished units before dividing.	17-24

The one habit that solves every problem in this module

Before you write a single figure, do this on your rough sheet. It takes forty seconds and it is the difference between a full-marks answer and a mess.



The units reconciliation. Write it for every process, every time.

THE THREE RATE FORMULAS — THIS IS ALL THE ARITHMETIC THERE IS

$$\text{Cost per unit (no loss)} = \frac{\text{Total cost of the process}}{\text{Units produced}}$$

$$\text{Cost per good unit (with loss)} = \frac{\text{Total cost} - \text{Scrap value of normal loss}}{\text{Input units} - \text{Normal loss units}}$$

Learn this one properly. Normal loss is removed from BOTH the top and the bottom. Abnormal loss and abnormal gain are then valued at this same rate — never at a different one.

$$\text{Cost per unit (part-finished stock)} = \frac{\text{Cost of that element}}{\text{Equivalent units of that element}}$$

Material, labour and overhead each get their own rate, because they are each at a different stage of completion.

Q1 Three processes, overhead on wages

Plain chain · p.8

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

There are three distinct processes A, B and C. During the four-week period 1,000 units are produced and the following information is available.

Particulars	Process A	Process B	Process C
Direct materials	2,000	1,000	—
Direct wages	1,500	700	800
Direct expenses	300	100	—

The indirect production cost were ₹4,500 and these are to be apportioned on the basis of direct wage cost. Prepare the process accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

Three things to notice, and they decide the whole answer:

- **“1,000 units are produced”** — one figure for all three processes. So there is **no loss anywhere**. 1,000 units go in, 1,000 come out of every process. That means no normal loss working, no abnormal loss — the easiest kind.
- **“apportioned on the basis of direct wage cost”** — the question hands you the basis. You do not have to guess it. Whenever a question gives you a lump overhead and a basis, your first working note is always the same: find the ratio, then split.
- **Process C has no materials and no direct expenses**. Do not panic and do not leave the row out — write a dash. An examiner wants to see that you looked and found nothing.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Add up the given basis across all processes (here: total direct wages).
2. Overhead rate = Total indirect cost ÷ Total basis. Keep it as a plain number or a percentage — whichever divides cleanly.
3. Multiply each process's basis by that rate. **Check the three pieces add back to the original lump sum** before you go on.
4. Open Process A. Debit its own four costs. Total it. That total transfers out to B.
5. Open Process B. **First debit is the transfer from A**, then B's own costs. Total, transfer to C.
6. Open Process C the same way. Its total goes to Finished Stock.
7. Divide each process total by 1,000 to show cost per unit. The last one is the answer the examiner is looking for.

WORKING NOTES

W1 Apportionment of indirect production cost

Total direct wages = 1,500 + 700 + 800 = **₹3,000**

Overhead rate = ₹4,500 ÷ ₹3,000 = **1.5 times the wages** (i.e. 150% of direct wages)

Process	Direct wages	× 1.5	Overhead absorbed
A	1,500	1,500 × 1.5	2,250
B	700	700 × 1.5	1,050
C	800	800 × 1.5	1,200
Total	3,000		4,500

Cross-check: 2,250 + 1,050 + 1,200 = 4,500 equals the indirect cost given. If this does not tally, stop and re-divide.

Process A Account

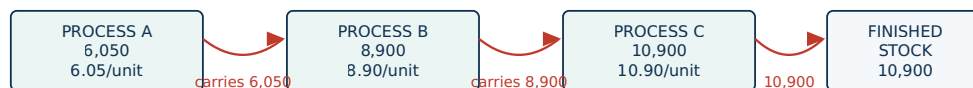
Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Direct materials	1,000	—	2,000	By Transfer to Process B 6,050 ÷ 1,000 units = ₹6.05 per unit	1,000	6.05	6,050
To Direct wages	—	—	1,500				
To Direct expenses	—	—	300				
To Indirect production cost W1: 1,500 × 1.5	—	—	2,250				
Total	1,000	—	6,050	Total	1,000	—	6,050

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A ← total of Process A a/c	1,000	6.05	6,050	By Transfer to Process C $8,900 \div 1,000 \text{ units} = ₹8.90 \text{ per unit}$	1,000	8.90	8,900
To Direct materials	—	—	1,000				
To Direct wages	—	—	700				
To Direct expenses	—	—	100				
To Indirect production cost W1: 700×1.5	—	—	1,050				
Total	1,000	—	8,900	Total	1,000	—	8,900

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B ← total of Process B a/c	1,000	8.90	8,900	By Transfer to Finished Stock $10,900 \div 1,000 = ₹10.90 \text{ per unit}$	1,000	10.90	10,900
To Direct materials	—	—	—				
To Direct wages	—	—	800				
To Direct expenses	—	—	—				
To Indirect production cost W1: 800×1.5	—	—	1,200				
Total	1,000	—	10,900	Total	1,000	—	10,900



Each box adds only its OWN cost; the arrow carries the whole running total forward.

The chain. Notice the cost per unit can only go up as you move right.

THE TRAP / WHERE MARKS ARE LOST

- Students write Process B's cost per unit as $2,850 \div 1,000 = ₹2.85$, using only B's own costs. Wrong. The unit leaving B has A's cost inside it too. **The transfer-in is a debit like any other.**
- Forgetting to show **units** in the account. Marks are given for the units column, and it is free — every figure is 1,000.
- Apportioning the overhead on the wrong basis (materials, or equally between three processes). The question said direct wage cost. Read it, underline it, use it.

FINAL ANSWER

Cost per unit — Process A	₹ 6.05
Cost per unit — Process B	₹ 8.90
Cost per unit — Process C (final)	₹ 10.90
Total cost of 1,000 units	₹ 10,900

State the final cost per unit in words as your closing line: "The cost of production is ₹10,900 for 1,000 units, i.e. ₹10.90 per unit."

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details show the cost of the three processes of manufacturing. The production of each process is passed on to the next till completion.

Particulars	Process A	Process B	Process C
Wages and materials	60,800	24,000	58,500
Works overheads	11,200	10,500	12,000
Production in units	72,000	75,000	96,000
Opening stock (units from preceding process on 1st July)	—	8,000	33,000
Closing stock (units from preceding process on 31st July)	—	2,000	11,000

READ THE QUESTION FIRST — WHAT TO SPOT

This problem looks like Q1 but it is not, and one phrase tells you so: “**units from preceding process**”.

- The opening and closing stock in Process B is **not half-finished work**. It is a pile of finished A-units waiting to be worked on. So it is valued at **A's rate**, not B's. Same for C — its stock is B-units, valued at B's rate.
- Because there is stock lying on both sides, the units B actually *consumed* are not the units it *received*. You must work the consumption out.
- “Production in units” is the **output** of that process. Divide by this figure to get the rate — never by the units consumed.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. **Process A first** — it has no stock, so rate = total cost ÷ production.
2. For B and C, find the units consumed: **Opening stock + Received from previous process – Closing stock**.
3. Value those consumed units at the **previous process's rate**. That is the transfer-in cost.
4. Add the process's own wages, materials and works overheads.
5. Rate for this process = that grand total ÷ **its own production in units**.
6. Now write the account. Debit side: opening stock, transfer-in, own costs. Credit side: closing stock (at the *previous* rate) and transfer-out (at *this* rate). The two sides must balance to the rupee.

WORKING NOTES

W1 Process A rate (no stock, so straight division)

Total cost = 60,800 + 11,200 = **₹72,000**

Rate = 72,000 ÷ 72,000 units = **₹1.00 per unit**

W2 Units consumed by Process B and Process C

Particulars	Process B	Process C
Opening stock of previous-process units	8,000	33,000
Add: Received from previous process	72,000	75,000
Units available	80,000	1,08,000

Particulars	Process B	Process C
Less: Closing stock of previous-process units	(2,000)	(11,000)
Units consumed in the process	78,000	97,000

Received by B = A's production 72,000. Received by C = B's production 75,000.

W3 Rate for Process B

Cost of units consumed = $78,000 \times ₹1.00 = 78,000$
Add own cost = $24,000 + 10,500 = 34,500$
Total = **₹1,12,500**
Rate = $1,12,500 \div 75,000$ units produced = **₹1.50 per unit**

W4 Rate for Process C

Cost of units consumed = $97,000 \times ₹1.50 = 1,45,500$
Add own cost = $58,500 + 12,000 = 70,500$
Total = **₹2,16,000**
Rate = $2,16,000 \div 96,000$ units produced = **₹2.25 per unit**

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Wages and materials	72,000	—	60,800	By Transfer to Process B W1: rate ₹1.00	72,000	1	72,000
To Works overheads	—	—	11,200				
Total	72,000	—	72,000	Total	72,000	—	72,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock 8,000 A-units @ ₹1.00 ← A's rate	8,000	1	8,000	By Closing stock 2,000 A-units @ ₹1.00 — still A's rate	2,000	1	2,000
To Transfer from Process A	72,000	1	72,000	By Transfer to Process C W3: 75,000 @ ₹1.50	75,000	1.50	1,12,500
To Wages and materials	—	—	24,000	(3,000 units consumed but not yielded — process shrinkage, no scrap value given)	3,000	—	—
To Works overheads	—	—	10,500				
Total	80,000	—	1,14,500	Total	80,000	—	1,14,500

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock <i>33,000 B-units @ ₹1.50 ← B's rate</i>	33,000	1.50	49,500	By Closing stock <i>11,000 B-units @ ₹1.50</i>	11,000	1.50	16,500
To Transfer from Process B	75,000	1.50	1,12,500	By Transfer to Finished Stock <i>W4: 96,000 @ ₹2.25</i>	96,000	2.25	2,16,000
To Wages and materials	—	—	58,500	<i>(1,000 units shrinkage)</i>	1,000	—	—
To Works overheads	—	—	12,000				
Total	1,08,000	—	2,32,500	Total	1,08,000	—	2,32,500



Why Process B's stock is valued at A's rate, not B's.

THE TRAP / WHERE MARKS ARE LOST

- Valuing B's opening/closing stock at B's own rate of ₹1.50. It has not been through B yet — it is worth ₹1.00. This single error breaks both sides of the account.
- Dividing by units consumed (78,000) instead of units produced (75,000). The rate must describe a unit that came out.
- Panicking when the units do not balance. Here 78,000 are consumed but only 75,000 emerge. That gap is real shrinkage. No scrap value is given, so it carries no cost — simply show it and move on. Do not invent a normal-loss working the question never asked for.

FINAL ANSWER

Cost per unit — Process A	₹ 1.00
Cost per unit — Process B	₹ 1.50
Cost per unit — Process C (final)	₹ 2.25
Cost transferred to Finished Stock	₹ 2,16,000

Both B and C accounts balance exactly — 1,14,500 and 2,32,500. Always show the totals; a balancing account is self-proving.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information relating to an article which undergoes three processes for manufacture, show the cost of each process and the cost per article.

Particulars	Process A	Process B	Process C
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250

The indirect expenses amounted to ₹21,250 and the number of articles produced during the month was 240 units.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same shape as Q1 — one output figure (240 articles), so **no losses**.
- **The basis for splitting the indirect expenses is NOT stated.** This is deliberate. The accepted convention when no basis is given is to apportion on **direct labour**. Say so in one line in your answer — “indirect expenses apportioned on the basis of direct labour in the absence of any other basis” — and you protect the marks.
- Test whether your assumed basis is right by checking the split comes out in round figures. Here labour gives exactly 25%. That is the examiner confirming your choice.
- Two answers are wanted: **cost of each process** and **cost per article**. Give both, clearly labelled.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the chosen basis — labour: 20,000 + 50,000 + 15,000.
2. Rate = 21,250 ÷ that total. Express as a percentage if it is clean.
3. Split, then cross-check the pieces add back to 21,250.
4. Build the three process accounts in a chain, transferring the running total.
5. Divide each process total by 240 to get cost per article at that stage.
6. Final line: cost per article = Process C total ÷ 240.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: direct labour)

Total labour = 20,000 + 50,000 + 15,000 = **₹85,000**

Rate = 21,250 ÷ 85,000 = 0.25 = **25% of direct labour**

Process	Labour	× 25%	Indirect expenses
A	20,000	20,000 × 0.25	5,000
B	50,000	50,000 × 0.25	12,500
C	15,000	15,000 × 0.25	3,750
Total	85,000		21,250

Cross-check: 5,000 + 12,500 + 3,750 = 21,250 — and every figure is a round number, which confirms labour was the intended basis.

Statement of Cost of Each Process and Cost per Article

Particulars	Process A	Process B	Process C
Cost of previous process (transferred in)	—	69,000	1,62,000
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250
Indirect expenses (W1)	5,000	12,500	3,750
Total cost of the process	69,000	1,62,000	2,37,000
Cost per article (÷ 240)	287.50	675.00	987.50

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials consumed	240	—	37,500	By Transfer to Process B $69,000 \div 240 = ₹287.50$	240	287.50	69,000
To Labour	—	—	20,000				
To Direct expenses	—	—	6,500				
To Indirect expenses W1	—	—	5,000				
Total	240	—	69,000	Total	240	—	69,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	240	287.50	69,000	By Transfer to Process C $1,62,000 \div 240 = ₹675.00$	240	675	1,62,000
To Materials consumed	—	—	12,500				
To Labour	—	—	50,000				
To Direct expenses	—	—	18,000				
To Indirect expenses W1	—	—	12,500				
Total	240	—	1,62,000	Total	240	—	1,62,000

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	240	675	1,62,000	By Transfer to Finished Stock $2,37,000 \div 240 = ₹987.50$	240	987.50	2,37,000
To Materials consumed	—	—	50,000				
To Labour	—	—	15,000				
To Direct expenses	—	—	6,250				
To Indirect expenses W1	—	—	3,750				
Total	240	—	2,37,000	Total	240	—	2,37,000

WHY THIS WORKS

The statement and the three accounts say exactly the same thing. If the question says “show the cost of each process” the statement alone is enough; if it says “prepare process accounts” you must draw the accounts. When in doubt give the statement first — it is faster to write and it makes the accounts almost automatic.

FINAL ANSWER

Cost of Process A ₹ 69,000

Cost of Process B ₹ 1,62,000

Cost of Process C ₹ 2,37,000

Cost per article ₹ 987.50

Q4 By-product and residue

Chain + by-product · p.8-9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A particular brand of chemical passed through three processes during the week ended 15th January. 600 tins were produced. The following information is disclosed by each process.

Particulars	Process A	Process B	Process C
Materials	4,000	2,000	1,500
Labour expenses	3,000	2,500	2,300
Direct expenses	6,000	200	500
Cost of tins	—	—	2,030
Packing charges	—	—	325

The indirect expenses were ₹1,600. The by-product is sold for ₹240 (Process B). The residue was sold for ₹125.50 (Process C).

Prepare the process A/c with cost of production of the product per tin.

READ THE QUESTION FIRST — WHAT TO SPOT

- Again 600 tins throughout — **no process loss**.
- Two new items: a **by-product sold for 240** and a **residue sold for 125.50**. Both are **deducted** from the cost of the process in which they arise. They are not income and they do not go to Profit & Loss — they reduce the cost that the good output has to carry.
- “Cost of tins” and “packing charges” belong to Process C only. They are ordinary debits — do not treat them as anything special.
- The indirect expense basis is again not stated, so apportion on labour and say so.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Apportion the indirect expenses on labour (state the assumption).
2. Build Process A normally.
3. In Process B, debit everything, then **credit the by-product sale of 240**. The amount that transfers to C is the total *after* that deduction.
4. In Process C, debit everything including tins and packing, then **credit the residue sale of 125.50**.
5. Cost of production = final balance of Process C. Divide by 600 tins.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: labour)

Total labour = 3,000 + 2,500 + 2,300 = ₹7,800

Rate = $1,600 \div 7,800 = 0.205128$ per rupee of labour

Process	Labour	Working	Indirect expenses
A	3,000	$3,000 \times 0.205128$	615.38
B	2,500	$2,500 \times 0.205128$	512.82
C	2,300	$2,300 \times 0.205128$	471.80
Total	7,800		1,600.00

The rate does not divide neatly here, so carry two decimals and force the last figure to make the total exactly 1,600. Write “difference of ₹0.01 adjusted in Process C” if you like — examiners accept it and it shows you checked.

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	600	—	4,000	By Transfer to Process B	600	22.69	13,615.38
To Labour expenses	—	—	3,000				
To Direct expenses	—	—	6,000				
To Indirect expenses W1	—	—	615.38				
Total	600	—	13,615.38	Total	600	—	13,615.38

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	600	22.69	13,615.38	By Sale of by-product deducted from cost, NOT income	—	—	240
To Materials	—	—	2,000	By Transfer to Process C $18,828.20 - 240 = 18,588.20$	600	30.98	18,588.20
To Labour expenses	—	—	2,500				
To Direct expenses	—	—	200				
To Indirect expenses W1	—	—	512.82				
Total	600	—	18,828.20	Total	600	—	18,828.20

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	600	30.98	18,588.20	By Sale of residue deducted from cost	—	—	125.50
To Materials	—	—	1,500	By Cost of production transferred $25,715.00 - 125.50 = 25,589.50$	600	42.65	25,589.50
To Labour expenses	—	—	2,300				
To Direct expenses	—	—	500				
To Cost of tins	—	—	2,030				
To Packing charges	—	—	325				
To Indirect expenses W1	—	—	471.80				
Total	600	—	25,715	Total	600	—	25,715

THE TRAP / WHERE MARKS ARE LOST

- Adding the by-product sale to the credit of Profit & Loss instead of deducting it inside the process. In a **by-product of small value** the standard treatment is to credit the process. That is what “(Process B)” in the question is telling you.
- Deducting the residue from Process B or from the total instead of from Process C. The question names the process for each — use it.
- Rounding the per-tin figure too early. Carry the paise through and round only at the last line.

FINAL ANSWER

Total cost of production ₹ **25,589.50**

Number of tins **600**

Cost of production per tin ₹ **42.65**

Closing line: “Cost of production is ₹25,589.50 for 600 tins, i.e. ₹42.65 per tin.”

Q5 Normal loss and abnormal loss

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare a Process Account and Abnormal Loss Account from the following information.

Particulars	
Input of raw material	1,000 units @ ₹20 per unit
Direct material	₹4,200
Direct wages	₹6,000
Production overheads	₹6,000
Actual output transferred to Process II	900 units
Normal loss	5%
Value of scrap per unit	₹8

READ THE QUESTION FIRST — WHAT TO SPOT

This is the first problem with a loss, and it is the template for Q5–Q9. Four figures decide everything:

- **Input 1,000 units.** Normal loss is a percentage *of input* — so 5% of 1,000 = 50 units. Not 5% of output.
- **Expected output = 1,000 – 50 = 950 units.** Write this down. It is the figure you compare the actual against.
- **Actual output is 900**, which is *less* than expected 950. So 50 units have gone missing beyond what was allowed — that is **abnormal loss**. If actual had been more than 950 it would have been abnormal gain.
- **Scrap is worth 8 per unit.** Normal loss units are sold as scrap, so they do carry a value — and that value is removed from the cost before you divide.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits — every cost that entered the process.
2. Normal loss in **units** = given % × input units. Normal loss in **rupees** = those units × scrap rate.
3. Expected output = Input – Normal loss units.
4. Compare actual output with expected output. Shortfall = abnormal loss; excess = abnormal gain.
5. **Cost per good unit** = (Total cost – Scrap value of normal loss) ÷ (Input units – Normal loss units). Show this fraction in full.
6. Value the transfer-out AND the abnormal loss at that **same** rate.
7. Write the process account — it must balance. Then open the Abnormal Loss account: debit the process value, credit the scrap realised, and the balance goes to Profit & Loss.

WORKING NOTES

W1 Total cost debited to the process

Raw material 1,000 × ₹20 = 20,000
Direct material = 4,200
Direct wages = 6,000
Production overheads = 6,000
Total = ₹36,200

W2 The units reconciliation

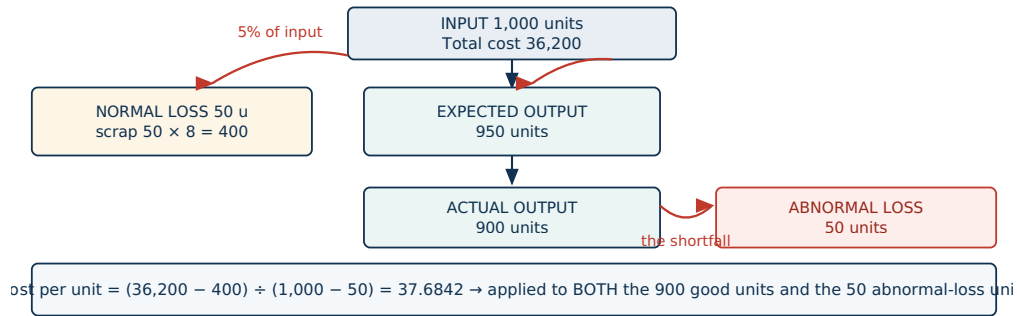
Normal loss = 5% of 1,000 = **50 units**, scrap value 50 × ₹8 = **₹400**
Expected output = 1,000 – 50 = **950 units**
Actual output = 900 units
Abnormal loss = 950 – 900 = **50 units**

W3 Cost per good unit — the formula that matters most

$$\text{₹36,200} - \text{₹400} \div 1,000 - 50 = \text{₹35,800} \div 950 = \text{₹37.6842 per unit}$$

Normal loss is taken out of the top (its scrap value) and out of the bottom (its units). That is how the cost of the normal loss gets loaded onto the good units.

Transferred out 900 × 37.6842 = **₹33,915.79**
Abnormal loss 50 × 37.6842 = **₹1,884.21**



Normal loss is expected and free. Abnormal loss is the shortfall against expectation.

Process I Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Raw material	1,000	20	20,000	By Normal loss 50 units × ₹8 scrap	50	8	400
To Direct material	—	—	4,200	By Abnormal loss W3: 50 × 37.6842	50	37.68	1,884.21
To Direct wages	—	—	6,000	By Transfer to Process II W3: 900 × 37.6842	900	37.68	33,915.79
To Production overheads	—	—	6,000				
Total	1,000	—	36,200	Total	1,000	—	36,200

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c ← from the process account	50	37.68	1,884.21	By Sale of scrap 50 units × ₹8 — abnormal loss units are also sold as scrap	50	8	400
Total	50	—	1,884.21	By Profit & Loss A/c balance — the real loss to the business	—	—	1,484.21
				Total	50	—	1,884.21

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c	50	8	400	By Cost Ledger Control A/c (cash from scrap)	50	8	400
Total	50	—	400	Total	50	—	400

THE TRAP / WHERE MARKS ARE LOST

- Dividing by 1,000 instead of 950. You would get ₹35.80 and every later figure would be wrong. **The denominator is always input minus normal loss units.**
- Forgetting to deduct the 400 scrap value from the numerator.
- Valuing the abnormal loss at the scrap rate of 8. No — abnormal loss is valued at the **good-unit rate**. It only realises 8 when sold, and the difference is the loss you take to Profit & Loss.
- Taking the whole 1,884.21 to Profit & Loss. Only the **net** figure of 1,484.21 goes there, after crediting the 400 the scrap actually fetched.

FINAL ANSWER

Cost per good unit **₹ 37.6842**

Transferred to Process II (900 units)	₹ 33,915.79
Abnormal loss (50 units, at cost)	₹ 1,884.21
Scrap realised on abnormal loss	₹ 400.00
Net abnormal loss charged to Profit & Loss	₹ 1,484.21

Note how the Process account, the Abnormal Loss account and the Normal Loss account all close cleanly. Show all three unless the question asks for fewer.

Q6 Loss, gain and neither — all in one

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The product of a company passes through 3 distinct processes. The following information is obtained from the accounts for the month ending January 31, 2024.

Particulars	Process A	Process B	Process C
Direct material	7,800	5,940	8,886
Direct wages	6,000	9,000	12,000
Production overheads	6,000	9,000	12,000

3,000 units @ ₹3 each were introduced to Process I. There was no stock of materials or work in progress. The output of each process passes directly to the next process and finally to Finished Stock A/c. The following additional data is obtained:

Process	Output	Normal loss %	Realisable value of scrap
Process 1	2,850	5%	2
Process 2	2,520	10%	4
Process 3	2,250	15%	5

Prepare Process Cost Account, Normal Loss Account and Abnormal Gain or Loss Account.

READ THE QUESTION FIRST — WHAT TO SPOT

This is **the** model problem of the module. It contains all three possibilities in one question, which is exactly why examiners like it.

- **Process 1 ends with no abnormal item at all** — actual output equals expected. Do not force an abnormal figure just because the other processes have one.
- **Process 2 has an abnormal loss** (output less than expected).
- **Process 3 has an abnormal gain** (output MORE than expected). An abnormal gain is entered on the **debit** side of the process account — the opposite side from a loss. This single fact is the most commonly dropped mark in the whole module.
- Normal loss % is always applied to **the units entering that process**, which is the previous process's output — not the original 3,000.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Work the processes strictly in order. Each one needs the previous one's output *and* its rate.
2. For each process: total the debits, compute normal loss units and scrap value, find expected output, compare with actual.
3. Rate = (Total cost – normal loss scrap) ÷ (Input units – normal loss units).

4. **Loss** → credit the process with the shortfall at that rate. **Gain** → debit the process with the excess at that rate.
5. Open one Normal Loss A/c for all three processes together. Debit each process's scrap value into it.
6. If there is an abnormal gain, the normal loss did not fully happen — so **credit Normal Loss A/c and debit Abnormal Gain A/c** with the gain units × that process's scrap rate.
7. Close Abnormal Loss and Abnormal Gain to Profit & Loss.

WORKING NOTES

W1 Process 1

Total cost = $(3,000 \times ₹3 = 9,000) + 7,800 + 6,000 + 6,000 = ₹28,800$

Normal loss = 5% of 3,000 = **150 units**; scrap $150 \times ₹2 = ₹300$

Expected output = $3,000 - 150 = 2,850$ units. Actual = 2,850 units.

Difference = NIL — no abnormal loss or gain in Process 1.

$$\text{Rate} = \frac{28,800 - 300}{3,000 - 150} = \frac{28,500}{2,850} = ₹10.00 \text{ per unit}$$

W2 Process 2

Total cost = $(2,850 \times ₹10 = 28,500) + 5,940 + 9,000 + 9,000 = ₹52,440$

Normal loss = 10% of **2,850** = **285 units**; scrap $285 \times ₹4 = ₹1,140$

Expected output = $2,850 - 285 = 2,565$ units. Actual = 2,520 units.

Abnormal LOSS = 2,565 - 2,520 = 45 units

$$\text{Rate} = \frac{52,440 - 1,140}{2,850 - 285} = \frac{51,300}{2,565} = ₹20.00 \text{ per unit}$$

Abnormal loss $45 \times 20 = ₹900$

Transfer to Process 3 $2,520 \times 20 = ₹50,400$

W3 Process 3

Total cost = $(2,520 \times ₹20 = 50,400) + 8,886 + 12,000 + 12,000 = ₹83,286$

Normal loss = 15% of **2,520** = **378 units**; scrap $378 \times ₹5 = ₹1,890$

Expected output = $2,520 - 378 = 2,142$ units. Actual = 2,250 units.

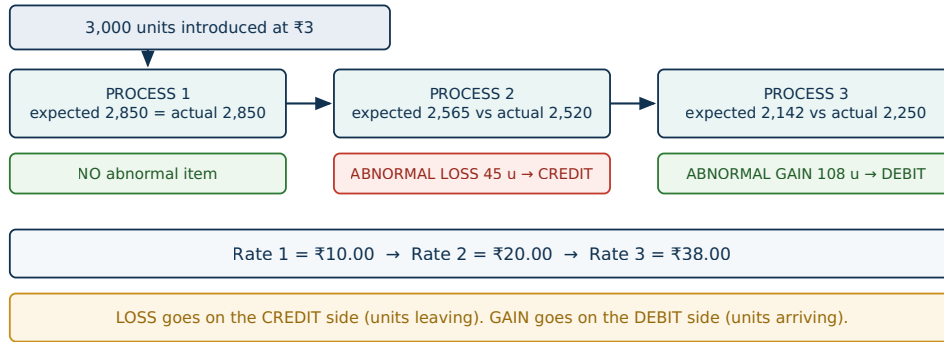
Abnormal GAIN = 2,250 - 2,142 = 108 units (actual is MORE than expected)

$$\text{Rate} = \frac{83,286 - 1,890}{2,520 - 378} = \frac{81,396}{2,142} = ₹38.00 \text{ per unit}$$

Note the rate is still built on EXPECTED output (2,142), not on the actual 2,250. The gain is then valued at this rate and added back.

Abnormal gain $108 \times 38 = ₹4,104$ (debit side)

Transfer to Finished Stock $2,250 \times 38 = ₹85,500$



One question, all three outcomes. Learn this picture and Q5 to Q9 all become the same problem.

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	3,000	3	9,000	By Normal loss $150 \text{ u} \times ₹2$	150	2	300
To Direct material	—	—	7,800	By Transfer to Process 2 $W1: 2,850 \times ₹10$	2,850	10	28,500
To Direct wages	—	—	6,000				
To Production overheads	—	—	6,000				
Total	3,000	—	28,800	Total	3,000	—	28,800

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 1	2,850	10	28,500	By Normal loss $285 \text{ u} \times ₹4$	285	4	1,140
To Direct material	—	—	5,940	By Abnormal loss $W2: 45 \times ₹20$	45	20	900
To Direct wages	—	—	9,000	By Transfer to Process 3 $W2: 2,520 \times ₹20$	2,520	20	50,400
To Production overheads	—	—	9,000				
Total	2,850	—	52,440	Total	2,850	—	52,440

Process 3 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 2	2,520	20	50,400	By Normal loss $378 \text{ u} \times ₹5$	378	5	1,890
To Direct material	—	—	8,886	By Transfer to Finished Stock $W3: 2,250 \times ₹38$	2,250	38	85,500
To Direct wages	—	—	12,000				
To Production overheads	—	—	12,000				
To Abnormal gain $W3: 108 \times ₹38$ — DEBIT side	108	38	4,104				
Total	2,628	—	87,390	Total	2,628	—	87,390

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	150	2	300	By Abnormal Gain A/c <i>108 u × ₹5 — the Process 3 normal loss that never happened</i>	108	5	540
To Process 2 A/c	285	4	1,140	By Cost Ledger Control A/c (scrap actually sold)	705	—	2,790
To Process 3 A/c	378	5	1,890				
Total	813	—	3,330	Total	813	—	3,330

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 2 A/c	45	20	900	By Cost Ledger Control A/c <i>45 u × ₹4 scrap</i>	45	4	180
Total	45	—	900	By Profit & Loss A/c <i>balance = the real loss</i>	—	—	720
				Total	45	—	900

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c <i>108 u × ₹5 — scrap income forgone</i>	108	5	540	By Process 3 A/c	108	38	4,104
To Profit & Loss A/c <i>balance = the real gain</i>	—	—	3,564				
Total	108	—	4,104	Total	108	—	4,104

THE TRAP / WHERE MARKS ARE LOST

- **Putting the abnormal gain on the credit side.** Think about it physically: extra units have *arrived* in the process, so they must be debited like any other input. If your Process 3 account will not balance, this is why.
- Applying the 15% normal loss to 3,000 instead of 2,520. Each percentage applies to that process's own input.
- Forgetting the entry **Abnormal Gain A/c → Normal Loss A/c 540**. Because 108 units did not perish, the company also did not earn scrap on them — so the gain must be reduced by $108 \times ₹5$. Without this the Normal Loss account will not close.
- Using the actual output 2,250 as the denominator for the rate. The denominator is always **input minus normal loss** = 2,142.

FINAL ANSWER

Cost per unit — Process 1	₹ 10.00
Cost per unit — Process 2	₹ 20.00
Cost per unit — Process 3	₹ 38.00
Abnormal loss (Process 2, 45 units)	₹ 900
Abnormal gain (Process 3, 108 units)	₹ 4,104
Transferred to Finished Stock (2,250 units)	₹ 85,500

Net loss to P&L on abnormal loss ₹ 720

Net gain to P&L on abnormal gain ₹ 3,564

Every one of the six accounts closes exactly. If yours do not, check the abnormal gain is on the debit side and that the 540 entry is present.

Q7 Abnormal gain only

Loss & gain · p.10

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Product A is obtained after it passes through three distinct processes. 2,000 kgs of material at ₹5 per kg were issued to Process 1, direct wages amounted to ₹900 and other production overheads incurred was ₹500. Normal loss is estimated at 10% of input. The wastage is sold at ₹3 per kg. The actual output is 1,850 kgs. Prepare Process 1 A/c, Normal Loss A/c and Abnormal Gain A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Only **Process 1** is asked for. The question mentions three processes to set the scene — ignore the other two. Answer exactly what is asked.
- Expected output = 2,000 – 200 = 1,800 kg. Actual is **1,850 kg — higher**. The question even names the account it wants: **Abnormal Gain A/c**. That is your confirmation before you calculate anything.
- Because there is a gain, remember the two-part consequence: debit the process with the gain at cost, and transfer the forgone scrap value to Normal Loss A/c.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits.
2. Normal loss = 10% of 2,000 = 200 kg; scrap value = 200 × 3.
3. Expected output 1,800 kg vs actual 1,850 kg → abnormal gain 50 kg.
4. Rate = (11,400 – 600) ÷ (2,000 – 200).
5. Debit the process with 50 kg at that rate; credit the transfer of 1,850 kg at that rate.
6. Normal Loss A/c: debit 200 kg at scrap rate; credit 50 kg to Abnormal Gain A/c; the rest is scrap sold.
7. Abnormal Gain A/c: debit the forgone scrap and the balance to P&L; credit the process figure.

WORKING NOTES

W1 Cost and units

Material 2,000 × ₹5 = 10,000

Direct wages = 900 | Production overheads = 500

Total cost = ₹11,400

Normal loss = 10% of 2,000 = **200 kg**; scrap 200 × ₹3 = **₹600**

Expected output = 1,800 kg | Actual output = 1,850 kg

Abnormal gain = 1,850 – 1,800 = 50 kg

$$\text{Rate} = \frac{11,400 - 600}{2,000 - 200} = \frac{10,800}{1,800} = \text{₹6.00 per kg}$$

Abnormal gain 50 × 6 = **₹300**

Transfer out 1,850 × 6 = **₹11,100**

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	5	10,000	By Normal loss 200 kg × ₹3	200	3	600
To Direct wages	—	—	900	By Transfer to Process 2 W1: 1,850 × ₹6	1,850	6	11,100
To Production overheads	—	—	500				
To Abnormal gain W1: 50 kg × ₹6	50	6	300				
Total	2,050	—	11,700	Total	2,050	—	11,700

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	200	3	600	By Abnormal Gain A/c 50 kg × ₹3 — loss that did not occur	50	3	150
Total	200	—	600	By Cost Ledger Control A/c (scrap sold)	150	3	450
				Total	200	—	600

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c scrap income forgone on 50 kg	50	3	150	By Process 1 A/c	50	6	300
To Profit & Loss A/c balance = net gain	—	—	150				
Total	50	—	300	Total	50	—	300

WHY THIS WORKS

The net gain reaching Profit & Loss is only ₹150, not ₹300. The process saved ₹300 of cost by producing 50 extra kg, but it also lost the ₹150 of scrap money those 50 kg would have fetched. Gain is always **cost saved minus scrap forgone**.

FINAL ANSWER

Cost per kg	₹ 6.00
Transferred out (1,850 kg)	₹ 11,100
Abnormal gain at cost (50 kg)	₹ 300
Scrap value forgone	₹ 150
Net abnormal gain to Profit & Loss	₹ 150

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through 2 distinct processes A and B. The normal wastage of each process is as follows: Process A – 3% of products entering the process; Process B – 5% of products entering the process.

The wastage of Process A was sold at 50 paise per unit and that of Process B at ₹1 per unit. 10,000 units were issued to Process A at ₹2 per unit. The other expenses were as follows:

Particulars	Process A	Process B
Materials	2,000	3,000
Wages	10,000	16,000
Overheads	2,100	2,375
Actual output (units)	9,500	9,100

Prepare process accounts and other ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- “3% of products entering the process” — so Process B’s 5% is on the 9,500 units it receives, not on the original 10,000.
- Process A: expected 9,700, actual 9,500 → **abnormal loss 200 units**.
- Process B: expected 9,025, actual 9,100 → **abnormal gain 75 units**.
- Two different scrap rates — 50 paise in A and ₹1 in B. Keep them apart; using one rate for both is a common slip.
- “and other ledger accounts” means: Normal Loss A/c, Abnormal Loss A/c and Abnormal Gain A/c. Draw all three.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Process A: total debits, normal loss 3% of 10,000, expected output, compare, rate, then value the loss and the transfer.
2. Carry A's output *and* A's rate into Process B as the transfer-in.
3. Process B: normal loss 5% of the units received (9,500), then the same sequence.
4. Build the three ledger accounts, remembering the Abnormal Gain → Normal Loss entry.

WORKING NOTES

W1 Process A

Total cost = $(10,000 \times ₹2 = 20,000) + 2,000 + 10,000 + 2,100 = ₹34,100$

Normal loss = 3% of 10,000 = **300 units**; scrap $300 \times ₹0.50 = ₹150$

Expected output 9,700 | Actual 9,500 → **abnormal loss 200 units**

$$\text{Rate} = \frac{34,100 - 150}{10,000 - 300} = \frac{33,950}{9,700} = ₹3.50 \text{ per unit}$$

Abnormal loss $200 \times 3.50 = ₹700$

Transfer to B $9,500 \times 3.50 = ₹33,250$

W2 Process B

Total cost = 33,250 + 3,000 + 16,000 + 2,375 = ₹54,625

Normal loss = 5% of 9,500 = 475 units; scrap 475 × ₹1 = ₹475

Expected output 9,025 | Actual 9,100 → **abnormal gain 75 units**

$$\text{Rate} = \frac{54,625 - 475}{9,500 - 475} = \frac{54,150}{9,025} = \text{₹6.00 per unit}$$

Abnormal gain 75 × 6 = ₹450 (debit side)

Transfer to Finished Stock 9,100 × 6 = ₹54,600

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	10,000	2	20,000	By Normal loss 300 u × ₹0.50	300	0.50	150
To Materials	—	—	2,000	By Abnormal loss W1: 200 × ₹3.50	200	3.50	700
To Wages	—	—	10,000	By Transfer to Process B W1: 9,500 × ₹3.50	9,500	3.50	33,250
To Overheads	—	—	2,100				
Total	10,000	—	34,100	Total	10,000	—	34,100

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	9,500	3.50	33,250	By Normal loss 475 u × ₹1	475	1	475
To Materials	—	—	3,000	By Transfer to Finished Stock W2: 9,100 × ₹6	9,100	6	54,600
To Wages	—	—	16,000				
To Overheads	—	—	2,375				
To Abnormal gain W2: 75 × ₹6 — DEBIT	75	6	450				
Total	9,575	—	55,075	Total	9,575	—	55,075

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	300	0.50	150	By Abnormal Gain A/c 75 u × ₹1 (B's scrap rate)	75	1	75
To Process B A/c	475	1	475	By Cost Ledger Control A/c (scrap sold)	700	—	550
Total	775	—	625	Total	775	—	625

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	200	3.50	700	By Cost Ledger Control A/c $200 \times ₹0.50$	200	0.50	100
Total	200	—	700	By Profit & Loss A/c	—	—	600
				Total	200	—	700

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c	75	1	75	By Process B A/c	75	6	450
To Profit & Loss A/c	—	—	375				
Total	75	—	450	Total	75	—	450

THE TRAP / WHERE MARKS ARE LOST

- Using ₹0.50 as the scrap rate in Process B. B's wastage sells for ₹1. Two processes, two rates.
- Taking 5% of 10,000 in Process B. It receives only 9,500.
- Netting the abnormal loss of Process A against the abnormal gain of Process B. They are separate accounts and separate P&L entries — never set one off against the other.

FINAL ANSWER

Cost per unit — Process A	₹ 3.50
Cost per unit — Process B	₹ 6.00
Abnormal loss (A, 200 units)	₹ 700 — net to P&L ₹ 600
Abnormal gain (B, 75 units)	₹ 450 — net to P&L ₹ 375
Transferred to Finished Stock (9,100 units)	₹ 54,600

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details are extracted from the costing records of an oil mill for the year ended 31st March, 2015. Purchase of 5,400 tonnes of coconut ₹2,20,000.

Particulars	Crushing	Refining	Finishing
Cost of labour	2,750	1,100	1,650
Electric power	660	396	264
Sundry material	110	2,200	—
Repairs to machinery	308	363	154
Steam	600	495	495
Factory expenses	1,452	726	242

Cost of sacks ₹8,250. 3,200 tonnes of crude oil were produced. 2,600 tonnes of oil produced by the refining process. 2,550 tonnes of refined oil were finished for delivery. Coconut sacks sold ₹440. 1,925 tonnes of coconut residue sold ₹12,100. Loss in weight in crushing 275 tonnes. 500 tonnes of by-products obtained from refining process ₹7,425.

Show the accounts for (a) coconut crushing process, (b) refining process, (c) finishing process, to arrive at the cost per tonne of each process and the total cost per tonne of the finished oil.

READ THE QUESTION FIRST — WHAT TO SPOT

A long question, but it is only Q1 with three deductions. The one thing that makes it easy is the **units reconciliation** — do it first and the whole answer falls out.

- **Crushing:** 5,400 t in. Out: crude oil 3,200 + residue 1,925 + loss in weight 275 = 5,400 . Perfect — no abnormal item.
- **Refining:** 3,200 t in. Out: refined oil 2,600 + by-product 500 = 3,100, so loss in weight = **100 t** (you have to work this out; the question does not give it).
- **Finishing:** 2,600 t in, 2,550 t out, so loss in weight = **50 t**.
- Three items are **credited** to their process because they are saleable: sacks 440 and residue 12,100 in crushing, by-product 7,425 in refining.
- Loss in weight is pure evaporation. It has **no scrap value**, so it carries no rupee figure — only tonnes. Show the tonnes and put a dash in the amount column.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Reconcile tonnes for each stage first. Work out any loss in weight the question has not given you.
2. Debit the process with the material coming in plus all its own expenses.
3. Credit any saleable output that is *not* the main product (sacks, residue, by-product) at its sale value.
4. Credit the loss in weight in tonnes only, amount nil.
5. The balance is the cost of the main output. Divide by its tonnage for cost per tonne.
6. Carry that balance forward as the debit of the next process.

WORKING NOTES

W1 Tonnage reconciliation — do this before anything else

Stage	In (t)	Main output (t)	Other output (t)	Loss in weight (t)
Crushing	5,400	3,200 crude oil	1,925 residue	275 (given)
Refining	3,200	2,600 refined oil	500 by-product	100 (derived)
Finishing	2,600	2,550 finished oil	—	50 (derived)

Refining loss = $3,200 - 2,600 - 500 = \mathbf{100 \text{ tonnes}}$

Finishing loss = $2,600 - 2,550 = \mathbf{50 \text{ tonnes}}$

W2 Cost per tonne at each stage

$$\text{Crushing} = \frac{2,21,590}{3,200 \text{ t}} = \mathbf{₹69.25}$$

$$\text{Refining} = \frac{2,19,445}{2,600 \text{ t}} = \mathbf{₹84.40}$$

$$\text{Finishing} = \frac{2,22,250}{2,550 \text{ t}} = \mathbf{₹87.16}$$

(a) Coconut Crushing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Coconut purchased	5,400	—	2,20,000	By Sale of sacks	—	—	440
To Cost of sacks	—	—	8,250	By Sale of coconut residue	1,925	—	12,100
To Cost of labour	—	—	2,750	By Loss in weight evaporation — tonnes only, no value	275	—	—
To Electric power	—	—	660	By Crude oil to Refining W2: 2,34,130 – 440 – 12,100	3,200	69.25	2,21,590
To Sundry material	—	—	110				
To Repairs to machinery	—	—	308				
To Steam	—	—	600				
To Factory expenses	—	—	1,452				
Total	5,400	—	2,34,130	Total	5,400	—	2,34,130

(b) Refining Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Crude oil from Crushing	3,200	69.25	2,21,590	By Sale of by-products	500	—	7,425
To Cost of labour	—	—	1,100	By Loss in weight W1: derived as 100 t	100	—	—
To Electric power	—	—	396	By Refined oil to Finishing W2: 2,26,870 – 7,425	2,600	84.40	2,19,445
To Sundry material	—	—	2,200				
To Repairs to machinery	—	—	363				
To Steam	—	—	495				
To Factory expenses	—	—	726				
Total	3,200	—	2,26,870	Total	3,200	—	2,26,870

(c) Finishing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Refined oil from Refining	2,600	84.40	2,19,445	By Loss in weight W1: derived as 50 t	50	—	—
To Cost of labour	—	—	1,650	By Finished oil (cost of delivery) W2: 2,22,250 ÷ 2,550 t	2,550	87.16	2,22,250
To Electric power	—	—	264				
To Repairs to machinery	—	—	154				
To Steam	—	—	495				
To Factory expenses	—	—	242				
Total	2,600	—	2,22,250	Total	2,600	—	2,22,250

THE TRAP / WHERE MARKS ARE LOST

- Treating the sacks, residue and by-product as income in Profit & Loss. They reduce the cost of the process that produced them.
- Giving the loss in weight a rupee value. It evaporated — there is nothing to sell. Tonnes only.
- Not deriving the refining and finishing losses. If you skip them the tonnage columns will not add up and the examiner will see it immediately.
- Dividing the crushing cost by 5,400 tonnes. The cost belongs to the 3,200 tonnes of crude oil that actually came out.

FINAL ANSWER

Cost per tonne — Crushing (crude oil)	₹ 69.25
Cost per tonne — Refining (refined oil)	₹ 84.40
Cost per tonne — Finishing (finished oil)	₹ 87.16
Total cost of 2,550 tonnes delivered	₹ 2,22,250

Closing line: “The total cost per tonne of the finished oil is ₹87.16.”